AMAZON – INTERVIEW EXPERIENCE

About the Company:

Amazon is one of the world's largest and most innovative multinational technology companies. With a strong customer-first approach, Amazon is renowned for revolutionizing online shopping through services like fast delivery, Amazon Prime, and Alexa-enabled smart devices. Its diverse business operations and commitment to innovation make it a dream workplace for professionals worldwide, offering unparalleled learning opportunities and a dynamic work environment.

Role: Support Engineer
Stipend : 50,000 + 400 USD per month

During the month of November, Amazon conducted a hiring drive for the role of Support Engineer. It was the second time after August that Amazon conducted a drive, having previously recruited candidates for the role of Software Development Engineer.

Overall Process:

Online Test, Technical Interview I, Technical Interview II

Online Test:

69 of us who applied were allowed to sit for the online test comprising 14 sections covering **DBMS, OOPS, Programming, Networking, Error Handling, Software Engineering, Data Visualization, and more.** It was all in MCQ format. We had one DSA question to solve (easy-level question).

Technical Interview I:

17 of us made it to Round 1, i.e., Technical Interview I.
The virtual interview happened on the AWS Chime platform, and I had two interviewers.

- **Initially, they asked me to introduce myself.**
- In my resume, I had projects in Machine Learning, Full Stack Development, and Front-End Development.

Question 2: *You have done projects in several domains. How did you manage this?*
I answered:
"Yes, sir, of course, we developed projects in several domains. The thing is, we would focus on a specific domain during each semester. For example, during our second semester, we focused on front-end development. In our fourth semester, we focused on back-end development and the database part. So, it was like a cumulative process, building on everything we learned. Therefore, it wasn't that hard."

- **Question 3:** *Explain your Machine Learning project.*
  (Note: In all three interviews I attended, the interviewers never skipped asking about the ML project. Since AI/ML is everywhere, be thorough with your ML project if you include one in your resume.)
  I explained my project, *Android Malware Detection System*, and the algorithm I used.
- **Question 4:** *What was the reason behind choosing this specific project? What was the problem statement you had in mind?*
  I answered with a real-life example I had faced.

They tested my knowledge in DBMS asking
How will you apply any condition in a query ?
What keyword will be used to fetch unique values from a column ?

Next, they moved on to **DSA**.
A live coding link was shared where the interviewer and I could view and edit on the same platform.

DSA Question 1: *How will you find the occurrence of a character in a given string?*
https://www.geeksforgeeks.org/program-count-occurrence-given-character-string/

I explained the approach using a hash map. They then asked me to code it (the logic function alone). Afterward, I dry-ran the code with an example and tested it against a test case.
Even though the time complexity was O(N), the interviewer asked me to reduce it to O(N/2). At that moment, I came up with the idea of using a two-pointer approach along with a hash map. The interviewer was satisfied.

- **Scenario-based Question:**
  What if I want the same alphabet in upper or lower case to be considered the same (e.g., 'a' and 'A')? How will you modify your code?
  I suggested converting all the characters in the string to a specific case (either upper or lower) before counting occurrences.

DSA Question 2: *Given an array, place the odd-indexed elements first and the even-indexed elements next.*
I gave an approach with O(N) complexity, typed the logic, and dry-ran it.

Finally, they asked me if I had any questions for them. I asked for feedback on my performance, and they provided some. The two interviewers were very encouraging and appreciated every answer I gave. The process was smooth, and the interview lasted around an hour.

Key Takeaways:

- Practice DSA.
- While coding, explain your approach.
- Take care of edge cases.

Technical Interview II:

9 of us made it to Round II (Technical Interview II).
Like Round 1, this interview happened on AWS Chime with a live coding link.

Introduction:
I introduced myself, and the interviewers introduced themselves.

DSA Questions:

1. Find the missing number.

https://leetcode.com/problems/missing-number/

2. Loop in a linked list.

https://leetcode.com/problems/linked-list-cycle/
I was asked to code and dry-run these with a test case. I solved and explained both questions.

Linux Commands:

1. *How will you find a specific word in a given text file?*
2. *What is the command used to edit a text file?*
3. One more similar question.
   I couldn’t answer all three correctly, but I attempted to think of and try something.

OS Questions:

1. What is a deadlock? What are the necessary conditions for a deadlock?

2. Explain CPU scheduling algorithms with examples.
   I took four processes, assumed their arrival time and burst time, and explained how SJF (Shortest Job First), SRJF (Shortest Remaining Job First), RR (Round Robin), PS (Priority Scheduling), and FCFS (First Come First Serve) would work.

Project Questions:
They asked me to explain a project close to my heart, the tech stack used, and where JavaScript was used in my project and why.

The interview lasted just half an hour and happened on **November 14th**.

Key Takeaways:

Be confident.
It’s not about answering all the questions right. Give what you can.

As I said , Virtual Interview happened on November 14th,

All my friends departed for our national industrial visit (IV). I stayed back in college to attend the virtual interview, while a friend of mine attended an in-person interview on campus the same day. After completing the interview, four of us somehow caught up with our friends in Bangalore and managed to join the IV.

On the last day of the IV (November 18th), I got the result:
Three of us, two from Computer Science and one from Information Technology, got selected.

I was one of the two Computer Science candidates.

A Personal Note:

I am not a pro coder or anything, and I come from a bio-math background in HSE. I joined college with no prior knowledge of what Computer Science actually was. Of course, I felt inferior at times, thinking, "Coding isn't meant for us." But I realized that Computer Science isn't just about typing and printing something on the screen. It's about understanding the things that happen behind the scenes. It's not merely about coding; it's about understanding the concepts.

Best of Luck!

Resources I Followed:

- **DSA:** Striver's SDE Sheet, Nikhil Lohia's and Striver's YouTube videos (Personal suggestion : Nikhil Lohia's explanations are top-notch).

https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/
https://www.youtube.com/@nikoo28/featured

- **OS, DBMS, OOPS, C++:** Career Ride YouTube channel. Their 20–25 minute videos cover every important topic. You can still explore those topics in-depth. https://www.youtube.com/results?search_query=careerride

- **CN, Linux:** GFG, Interview Bit, and similar online platforms.

Feel Free to reach out if you have any queries

Email : tharaniarumugam514@gmail.com
Phone: 6382033914

LinkedIn : https://www.linkedin.com/in/tharani-arumugam?utm_source=share&utm_campaign=share_via&utm_content=profile&utm_medium=android_app